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PERFORMANCE SPECIFICATION

CAPACITOR, VARIABLE, CERAMIC DIELECTRIC, GENERAL SPECIFICATION FOR

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for ceramic dielectric; variable capacitors intended for operation over a temperature range of -55°C to +85°C, for use in circuits where periodic adjustments are required (see 6.1).

1.2 Classification.

1.2.1 Part or Identifying Number (PIN). The PIN is in the following form, and as specified (see 3.1 and 6.2).

CV11	A	070
Style (see 1.2.1.1)	Characteristic (see 1.2.1.2)	Capacitance (see 1.2.1.3)

1.2.1.1 Style. The style is identified by the two-letter symbol “CV” followed by a two-digit number; the letters identify ceramic dielectric, variable capacitors and the number identifies the shape and dimensions of the capacitor.

1.2.1.2 Characteristic. The characteristic is identified by a single letter in accordance with table I.

1.2.1.3 Capacitance. The nominal maximum rated capacitance value expressed in picofarads (pF) is identified by a three-digit number; the first two digits represent significant figures and the last digit specified the number of zeros to follow. When fractional values are required, the letter “R” will be used to indicate the decimal point. For example: 2R5 indicates 2.5 pF.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus, OH 43218-3990, or emailed to capacitorfilter@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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TABLE I. Characteristics.

Symbol	Capacitance change from value at +25˚C (percent)					
	At -55˚C		At +85˚C		At +125˚C	
	Min.	Max.	Min.	Max.	Min.	Max.
A[3]	-4.5	+2.0	-2.5	+2.0	-4.2	+3.4
B	-1.0	+3.5	-2.5	-0.5	-4.2	-0.8
C	-1.0	+6.5	-4.0	-1.0	-6.7	-1.7
D	+1.5	+8.2	-5.0	-1.5	-8.5	-2.5
E	+3.0	+14.0	-10.0	-3.0	-15.0	-9.0
G	0.0	+14.0	-8.0	-3.0	-14.0	-5.0

	Temperature coefficient -55˚C to +85˚C (ppm/˚C)
F	+100 ± 150
H	+100 ± 600
J	-300 ± 300
K	-450 ± 300
L	+50 ± 150
M	-100 ± 500
N	-300 ± 800
P	-400 ± 500
Q	-1500 ± 900

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-81/1	-	Capacitors, Variable, Ceramic Dielectric, Style CV11
MIL-PRF-81/4	-	Capacitors, Variable, Ceramic Dielectric, Styles CV31 and CV32
MIL-PRF-81/6	-	Capacitors, Variable, Ceramic Dielectric, Styles CV35 and CV36

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Electronic and Electrical Component Parts
MIL-STD-202-106	-	Method 106, Moisture Resistance
MIL-STD-202-204	-	Method 204, Vibration, High Frequency
MIL-STD-202-213	-	Method 213, Shock (Specified Pulse)
MIL-STD-202-215	-	Method 215, Resistance to Solvents
MIL-STD-202-301	-	Method 301, Dielectric Withstanding Voltage
MIL-STD-202-302	-	Method 302, Insulation Resistance
MIL-STD-202-305	-	Method 305, Capacitance

(Copies of these documents are available online at https://quicksearch.dla.mil/ .)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Qualification is not required.

3.3 Material. The material shall be as specified herein; however, when a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4 Interface and physical dimensions. Capacitors shall meet the interface, and physical dimensions specified (see 3.1).

3.4.1 Pure tin. The use of pure tin as an underplate or final finish is prohibited both internally and externally. Tin content of capacitor components and solder shall not exceed 97 percent by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.7).

3.5 Dielectric withstanding voltage. When measured as specified in 4.6.2, capacitors shall withstand the direct current (dc) potential without damage, breakdown, or flashover.

3.6 Insulation resistance. When measured as specified in 4.6.3, the insulation resistance of the capacitors shall be not less than 10,000 megohms.

3.7 Capacitance. When measured as specified in 4.6.4, the minimum capacitance shall be no greater than the minimum value specified and the maximum capacitance shall be not less than the maximum value specified, and no greater than 50 percent above the maximum value specified (see 3.1).

3.8 Dissipation factor. When measured as specified in 4.6.5, the dissipation factor of capacitors shall be not more than 0.2 percent.

3.9 Temperature coefficient. When measured as specified in 4.6.6, capacitance change measurements shall be within the limits specified in table I for the applicable characteristic.

3.10 Terminal strength. When measured as specified in 4.6.7, there shall be no fracture, loosening of parts, or other mechanical failures.

3.11 Torque. When measured as specified in 4.6.8, the torque values shall be within the limits specified for individual capacitor styles (see 3.1).

3.12 Fatigue. When measured as specified in 4.6.9, capacitors shall meet the following requirements:

a. Capacitance: Shall change not more than ± 12 percent or 0.75 pF, whichever is greater, from the initial value obtained when measured as specified in 4.6.9.

b. Torque: As specified for individual capacitor styles (see 3.1).

3.13 Life. When tested as specified in 4.6.10, capacitors shall meet the following requirements:

a. Insulation resistance: As specified in 3.6.

b. Capacitance: Unless otherwise specified, shall change not more than ± 5 percent or 0.5 pF, whichever is greater, from the initial value obtained when set and measured as specified in 4.6.10.

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3.14 Shock, specified pulse. When capacitors are tested as specified in 4.6.11, there shall be no intermittent contacts of 0.5 millisecond (ms) or greater duration or arcing or other indication of breakdown, nor shall there be any open or short circuiting or evidence of mechanical damage.

3.15 Vibration, high frequency. When capacitors are tested as specified in 4.6.12, there shall be no intermittent contacts of 0.5 ms or greater duration or arcing or other indication of breakdown, nor shall there be any open or short circuiting or evidence of mechanical damage. Capacitors shall also meet the following requirements:

a. Capacitance: Shall change not more than ± 2 percent or 0.5 pF (0.25 pF for values less than 10 pF), whichever is greater, from the initial value obtained when measured as specified in 4.6.4.

b. Dissipation factor: Not more than 0.2 percent.

c. Dielectric withstanding voltage: As specified in 3.5.

d. Insulation resistance: As specified in 3.6.

3.16 Moisture resistance. When measured as specified in 4.6.13, capacitors shall meet the following requirements:

a. Visual inspection: There shall be no mechanical damage.

b. Insulation resistance: As specified in 3.6.

c. Capacitance: Unless otherwise specified (see 3.1), shall change not more than ± 5 percent or 0.5 pF, whichever is greater, from the initial value obtained when set and measured as specified in 4.6.4.

d. Dissipation factor: Shall not be more than 0.5 percent.

3.17 Marking. Each capacitor shall be legibly marked with smear-resistant ink that will withstand the environmental tests specified herein and shall be marked with the PIN, and the manufacturer's name or symbol. The marking shall not be placed on the mounting surface. There shall be no space between the symbols which comprise the PIN. If lack of space requires it, the PIN may appear on two lines. In this event, the PIN shall be divided between the style and the characteristic symbols and shall appear on two lines as shown in the following example:

CV11
A070

Other markings which in any way interfere with, obscure, or confuse those specified herein, are prohibited. Marking shall remain legible after all tests.

3.18 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.19 Workmanship. Capacitors shall be manufactured and processed in such a manner as to be uniform in quality and be free from other defects that will affect life, serviceability, or appearance.

3.19.1 Soldering. All excess flux or solder shall be removed. Electrical connections shall be electrically continuous after soldering.

3.19.2 Finish of terminals. The terminals shall be so coated that soldering can be accomplished readily.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Conformance inspection (see 4.4).

b. Group C inspection (see 4.5).

4.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the "GENERAL REQUIREMENTS" of MIL-STD-202. Accuracy of all test voltage measurements shall be within ± 2.0 percent of the specified voltage.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A, group B, and group C inspections.

4.4.1.1 Certification. The acquiring activity, at its discretion, may accept a certificate of compliance with group C requirements in lieu of performing group C tests (see 6.2c). NOTE: Group C testing includes a 1,000-hour life test which may extend the processing time by 45 days or more.

4.4.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table II, in the order shown.

4.4.2.1 Sampling plan. The sampling plan shall be as specified in table II.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Method paragraph	Sampling procedure
Subgroup 1			
Visual and mechanical inspection: design, construction, and physical dimensions	3.1 and 3.4	4.6.1	13 samples
Marking	3.17		0 failures
Workmanship	3.19 to 3.19.2 inclusive		
Subgroup 2			
Dielectric withstanding voltage	3.5	4.6.2	
Insulation resistance	3.6	4.6.3	13 samples
Capacitance	3.7	4.6.4	0 failures
Dissipation factor	3.8	4.6.5	

4.4.3 Group B inspection. Group B inspection shall consist of the tests specified in table III, in the order shown. Parts may not be shipped until representative samples of the lot have successfully completed the group B tests. In the event of 1 or more failures, the lot shall be rejected.

4.4.3.1 Sampling plan. The sampling plan shall be as specified in table III.

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TABLE III. Group B inspection.

Inspection	Requirement paragraph	Method paragraph	Sampling procedure
Terminal strength Torque Temperature coefficient	3.10 3.11 3.9	4.6.7 4.6.8 4.6.6	5 samples 0 failures

4.4.3.2 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.

4.4.3.3 Rejected lots. The rejected lot shall be segregated from new lots and those that have passed inspection. Rejected lots shall be 100 percent reworked or scrapped. The rejected lot may be rescreened and the defects removed. The lot may then be resubmitted to the sample plan. If one or more defects of the same type are found in this second sample, the lot is rejected and shall not be supplied to this specification. If another defect of a different type is found in the second sample, a rescreen for that defect is also permitted.

4.5 Group C inspection. Group C inspection shall consist of the tests specified in table IV, in the order shown. The number of sample units to be inspected shall be as specified in table IV. In the event of failures in excess of those allowed in table IV, the lot shall be rejected. Parts may not be shipped until representative samples of the lot have successfully completed the group C tests.

4.5.1 Rejected lots. Rejected lots shall not be supplied to this specification.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Method paragraph	Number of sample units to be inspected	Number of failures allowed
Subgroup 1				
Fatigue Life	3.12 3.13	4.6.9 4.6.10	12	1
Subgroup 2				
Shock, specified pulse Vibration, high frequency Moisture resistance	3.14 3.15 3.16	4.6.11 4.6.12 4.6.13	18	1

4.6 Methods of inspection.

4.6.1 Visual and mechanical examination. Capacitors shall be examined to verify that the materials, design, construction, physical dimension, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.17, and 3.19).

4.6.2 Dielectric withstanding voltage (see 3.5).

4.6.2.1 Terminal to terminal. Capacitors shall be tested in accordance with MIL-STD-202-301. The following details shall apply:

a. Magnitude of test voltage: 2.2 times applicable rated dc voltage.

b. Duration of application of test voltage: 3 seconds ± 2 seconds.

c. Points of application of test voltage: Between terminals.

d. Limiting value of surge current: 50 milliamps.

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4.6.2.2 Terminal to mounting. Capacitors shall be tested in accordance with 4.6.2.1 with the following exceptions:

a. Capacitors shall be mounted by normal mounting means on a metal plate and the terminals shall be electrically connected. (Resilient spacer referenced in 6.4 shall not be used for this test.)

b. In lieu of 4.6.2.1c use the points of application of test voltage: Between terminals and metal plate.

c. Styles CV31 and CV32 do not require this test.

4.6.3 Insulation resistance (see 3.6). Capacitors shall be tested in accordance with MIL-STD-202-302. The following details shall apply:

a. Test condition: Rated voltage, but not to exceed 100 volts.

b. Special conditions: If failure occurs at a relative humidity above 60 percent, the insulation resistance may be measured again at a relative humidity of 50 percent ± 5 percent.

c. Point of measurement: Terminal to terminal.

d. Electrification time: Shall not exceed 1 minute.

4.6.4 Capacitance (see 3.7). Capacitors shall be tested in accordance with MIL-STD-202-305. The following details shall apply:

a. Test frequency: 1 MHz ± 100 kHz.

b. Limit of accuracy: ± 2 percent or 0.5 pF, whichever is greater.

4.6.5 Dissipation factor (see 3.8). The dissipation factor of capacitors shall be measured at both maximum and minimum capacitance setting at a frequency of 1 MHz ± 100 kHz. The accuracy of measurement shall be within ± 10 percent of the limit (see 3.8). The voltage applied shall be less than 30 volts peak.

4.6.6 Temperature coefficient (see 3.9). Capacitance-change measurement at 80 percent to 90 percent of maximum rated capacitance setting shall be made at the temperature specified in table V and at a frequency between 0.1 MHz and 1.2 MHz. Each measurement shall be made after the capacitor has reached thermal stability.

TABLE V. Capacitance-change measurements.

Sequence	Temperature
Step 1 Start at	+25˚C ±2˚C
Step 2 Reduce to	-55˚C ±2˚C
Step 3 Return to	+25˚C ±2˚C
Step 4 Raise to	+85˚C ±2˚C
Step 5 Return to	+25˚C ±2˚C

4.6.7 Terminal strength (see 3.10). A 4-pound force shall be applied between each terminal and the base. The force shall be applied parallel to the terminal extension, and shall be increased gradually to 4 pounds and held at that value for not less than 10 seconds.

4.6.8 Torque (see 3.11). The torque required to start and maintain rotation of the rotor shall be measured by a gradually applied force sufficient to turn the rotor through at least one complete rotation.

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4.6.9 Fatigue (see 3.12). Capacitance, at maximum capacitance setting, shall be measured as specified in 4.6.5. The rotors of the capacitors shall be continuously rotated through the complete range (360 degrees) in alternating directions at a rate of approximately 20 cycles per minute for a period of 5 minutes. The driving mechanism used in this test shall include a flexible coupling to prevent development of excessive pressures caused by misalignment of the driving shaft and setting, and the torque shall be measured as specified in 4.6.8, respectively.

4.6.10 Life (see 3.13). Capacitance shall be set at 80 percent to 90 percent of maximum capacitance and measured as specified in 4.6.5. A dc potential of 50 percent rated voltage upon which a peak alternating potential of 50 percent rated voltage at a frequency of 100 Hz or less is superimposed, shall then be applied to the capacitors for 1,000 hours ± 12 hours at a temperature of +85˚C ± 2˚C. Upon completion of the test, capacitors shall be returned to standard test conditions without disturbing the initial capacitance setting. Insulation resistance and capacitance shall be measured as specified in 4.6.3 and 4.6.4, respectively.

4.6.11 Shock, specified pulse (see 3.14). Capacitors shall be tested in accordance with MIL-STD-202-213. The following details and exceptions shall apply:

a. Special mounting means: Capacitors shall be rigidly mounted on a mounting fixture by the normal mounting means.

b. Number and direction of applied shocks: Thirty shocks, ten blows in each of three mutually perpendicular planes (five in each of two directions). The test planes shall be perpendicular to the normal mounting axis, perpendicular to the mounting axis with the capacitor body rotated 90 degrees, and parallel to the mounting axis.

c. Test-condition letter: I.

d. Electrical-load conditions: During the test, a potential of 125 percent of the rated dc voltage (see 3.1) shall be applied between the terminals of the capacitor.

e. Measurements during shock: During the test, observations shall be made to determine intermittent contacts of 0.5 ms or greater duration or arcing or other indication of any open or short circuiting or evidence of mechanical damage.

f. Examinations after test: Capacitors shall be visually examined for evidence of breakdown, arcing, and mechanical damage.

4.6.12 Vibration, high frequency (see 3.15). Capacitors shall be tested in accordance with MIL-STD-202-204. The following details and exception shall apply:

a. Mounting of specimens: Capacitors shall be mounted as specified in 4.6.11a. The mounting fixture shall be so constructed as to preclude any resonance within the test range. An inspection of the mounting fixture shall be made on a vibrator. If any resonant frequencies are observed, adequate steps must be taken to damp the structure.

b. Electrical-load conditions: As specified in 4.6.11d.

c. Test condition letter: B.

d. Measurements during vibration: During the last cycle in each direction and electrical measurement shall be made to determine intermittent contacts of 0.5 ms or greater duration or open or short circuiting.

e. Examination after vibration: Capacitors shall be visually examined for evidence of mechanical damage.

f. Final measurements: After the final vibration cycle, the initial capacitance setting shall not be disturbed. Dielectric withstanding voltage, insulation resistance, capacitance, and dissipation factor shall be measured as specified in 4.6.2, 4.6.3, 4.6.4, and 4.6.5, respectively.

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4.6.13 Moisture resistance (see 3.16). Capacitors shall be tested in accordance with MIL-STD-202-106. The following details shall apply:

a. Mounting: Securely fastened by normal mounting means.

b. Initial measurements: Capacitance shall be set at 80 percent to 90 percent of maximum rated capacitance and measured as specified in 4.6.4. Capacitance measurements shall be made not less than 1 hour nor more than 4 hours before test initiation.

c. Polarization and load: Not applicable.

d. Final measurements: After the final cycle, the initial capacitance setting shall not be disturbed. Insulation resistance, dissipation factor, and capacitance shall be measured as specified in 4.6.3, 4.6.4, and 4.6.5, respectively.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Capacitors covered by this specification are not intended for circuits where periodic adjustment is objectionable. Capacitors covered by this specification are unique due to the fact that these devices must be able to operate satisfactorily in high reliability military systems under the following demanding conditions: 15Gs of high frequency vibration, 100Gs of shock (specified pulse), and wide temperature fluctuations. These capacitors also offer very high reliability that is verified under a qualification system. Commercial components are not designed to withstand these military environmental conditions.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification, applicable specification sheet, and the complete PIN (see 1.2.1 and 3.1).

b. Packaging requirements (see 5.1).

c. Whether the manufacturer performs the group C inspections or provides a certificate of compliance with group C requirements (see 4.4.1.1).

6.3 Qualification. Qualification is not required.

6.4 Mounting. It is recommended that a resilient spacer between the unit and mounting surface be used to avoid possible cracking or chipping of the ceramic mounting base.

6.5 Identification of maximum capacitance. For style CV11, the approximate maximum capacitance value is obtained by rotating the plated area of the disk (rotor) until it coincides as nearly as possible with the area of the solder button terminal of the stator.

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6.6 Subject term (key word) listing.

Capacitance
Dissipation factor

6.7 Tin whisker growth. The use of alloys with a tin content greater than 97 percent, by mass may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture, and can develop under typical operating conditions on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead by mass have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

6.8 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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